Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/30/2025
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Project Title:	HCAAM VNSCOR: Enhancing Situation Awareness of Automated Procedures Using Adaptive Multimodal Augmented Reality Displays		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	05/15/2019	**End Date:**	06/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2025 per A. Beitman/JSC (Ed., 3/6/25) NOTE: End date changed to 03/31/2025 per A. Beitman/JSC (Ed., 9/19/24) NOTE: End date changed to 12/31/2024 per A. Beitman/JSC (Ed., 1/20/23) NOTE: End date changed to 5/14/2023 per S. Huppman/HRP and NSSC information (Ed., 3/3/2020)		
Key Personnel Changes/Previous PI:	March 2025 report: There are no key personnel changes.		
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Grant/Contract No.:	80NSSC19K0667		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR).
Future deep space missions will present new challenges for crew, and increased risks to human performance due to the stress, fatigue, radiation exposure, and isolation that characterizes these missions. In addition, crew will no longer be able to depend on timely support from Mission Control due to distance from the Earth, but will have to work autonomously, while maintaining high performance. Mission Controllers may not be available to answer questions, check system status, assist with procedures, monitor for errors, or troubleshoot problems. Greater crew autonomy will increase dependence on automated systems, and design of these automated systems must be driven by sound human-system integration standards and guidelines in order to ensure mission success. Historically, crew have had very limited dependence on automated systems, thus crew will be faced with a new way of working that may put situation awareness (SA) at risk. We must develop methods for promoting good situation awareness in the automated systems that will most certainly be part of future deep space vehicles and habitats.

Procedure automation is a promising technology for reducing crew workload. We define procedure automation as technology that automates the selection or execution of procedural tasks. Structuring the work of automation according to human procedures should improve the transparency of automation actions. This approach provides a means for establishing common ground about ongoing tasks to improve operator understanding of automation behavior.

New technologies such as adaptive, multimodal, augmented reality displays can offer the benefits of information presentation tailored to meet the needs of each crewmember, taking into consideration the current state of that crewmember (e.g., sleep-deprived, high workload), as well as the current state of his/her environment and ongoing activities (e.g., emergency situation, time-critical operations).

We combine technology for procedure automation with technology for augmented reality multi-modal (ARMM) user interfaces using Microsoft Hololens head-mounted display to provide a virtual task assistant to assist crew in performing procedural work. This virtual task assistant identifies procedures to perform, aids performing actions in crew procedures, and summarizes actions taken by the human-automation team to assist crew in preparing for tasks and taking over tasks from other team members.

Four studies were conducted to evaluate the effects of a virtual task assistant combining procedure automation with augmented reality multi-modal (AARM) user interfaces on human task performance. These studies will achieve the following aims:

Aim. Determine best methods to improve situation awareness and improve crew autonomy when using a virtual task assistant to prepare for and perform manual maintenance.

Aim. Determine best practices to improve crew performance when using virtual task assistant to aid in task transitions (task handovers, automation interruption).

Aim. Determine best methods for display of procedural work to improve crew autonomy and satisfaction

The proposed work addresses a number of gaps in the Human Research Program Human Factors and Behavioral Performance risks. This project will provide guidelines for designing effective human-automation systems and evaluate human-automation performance for exemplar procedure automation systems. This project also will provide guidance for the application of multi-modal and adaptive displays and control to Human-Computer Interaction (HCI) design for long duration operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Technologies for virtual task assistance are increasingly available in everyday life. One of the most common is voice enabled assistance, like Siri and Alexa, that aid some activities of daily living. And augmented and virtual reality technologies are becoming mainstream, with the introduction of new devices such as Microsoft HoloLens 2 and Apple's Vision Pro.
The Virtual Intelligent Task Assistant (VITA) project is leveraging augmented reality platforms and toolkits for augmented reality apps to develop a virtual task assistant that can be used to assist users with procedural task work on the job. Our technical approach is innovative in that new procedural tasks can be supported without custom software development. Our experimental research is distinguished by investigating effective task assistance for maintenance or assembly tasks where hands-free operation of task assistance is beneficial. For the first study we investigated best techniques for using augmented reality task assistance when assembling small rovers that are held in the hands during assembly. Subsequent studies investigate other aspects of using augmented reality for assembly tasks, including use of 3D images in procedures and virtual notification for task milestones and transitions.

This technology and associated research findings have potential benefit to NASA for the assembly, maintenance, and repair of aircraft, spacecraft, habitats, and robotics. This technology and associated research findings also have broader potential benefit for any organization performing assembly and maintenance procedural work. This includes assembly and maintenance of drilling equipment for the oil and gas industry, equipment used in chemical processing plants, and maintenance and repair of commercial aircraft.

I. VITA Study in HERA Campaign 7
The aim of the Virtual Intelligent Task Assistant (VITA) study in NASA Human Exploration Research Analog (HERA) Campaign 7 is to determine the best methods for the display of procedural work to improve crew autonomy and satisfaction. Specifically, this study compares crew performance and satisfaction on rover assembly tasks similar to those used in HERA Campaign 6 but with procedures that vary the level of detail in text instruction and the visual aids, to improve support for flexible execution of actions. These differences are illustrated in existing VITA procedures for rover assembly and disassembly.

A new version of the VITA task assistant was developed for use in HERA Campaign 7. The new task assistant is an augmented reality (AR) application developed in Unity. This application runs on the augmented reality headset Microsoft's HoloLens 2, instead of the HoloLens 1 used in HERA Campaign 6. The VITA augmented reality display was redesigned, including changing from gaze interaction used in HERA C6 to voice interaction for HERA C7. To

address our aim of the best methods for display of procedure work, we investigate procedures that vary the visual aid and the level of detail in text. Specifically, we compare performance with figures displaying Photographs to 3D models with diagrams in figures. We also look at the level of detail in text instructions, comparing the procedure with and without more detailed text instructions.

The VITA study in HERA Campaign 7 had a total of 16 participants. Participants were selected by HERA to be astronaut-like. Participants were trained to use VITA prior to each HERA mission. Data collection includes situation awareness, workload, usability, and task timing at the end of each session. A survey was administered at the end of each session (called the end of condition survey) and after the final VITA session (called the end of mission survey). These surveys include questions about user satisfaction and perception of autonomy when using different procedure styles will be added to the end of the mission survey.

Since the annual report was submitted in March 2025, the analysis of data collected during HERA C7 was completed. Findings are summarized on human performance using the different visual aids during the VITA study in HERA C7. We also summarize findings of our investigation into procedure style and crew sense of work autonomy.

Visual Aids in AR procedures

Augmented reality interfaces afford the opportunity to situate virtual overlays relative to the real world. A common technique is to use image recognition software like Vuforia https:// to locate objects in the world, then overlay virtual cues on the view of the object to highlight or annotate it. We found that such an approach did not work well for our rover assembly task. One issue is that the rover has many small parts that can be difficult to distinguish and track with software like Vuforia. Small components could also be occluded by the annotation. Another issue is that the rover's appearance changed with every assembly or disassembly action taken, making it challenging to capture and recognize all the possible appearances of the real object during the assembly or disassembly task. For such applications, we situated cues relative to a 3D model of the object. In HERA C7, we investigate the use of 3D models for small rover assembly, and compare performance with 3D models to performance with Photographs. This investigation of visual aids addresses our aims of the best methods for the display of procedural work as well as support for hands-free assembly and maintenance.

There is a significant reduction in workload when using 3D models for the first assembly session instead of Photographs. Workload is low (Median=3) and the same; however, for both conditions in the second assembly session. A Wilcoxon signed-rank test was done for median workload in the first session of both visual aids. Workload was higher for participants during the first rover assembly when using Photographs (Median=6) for a visual aid than for the 3D model (Median=3.5). This workload difference is significant.

This difference indicates that the 3D model may be easier to learn than the Photographs, when the task is unfamiliar to the user. Once the user practices with the visual aid, these workload differences disappear.

In addition to the effects of these visual aids on workload, some workload differences could be due to the task differences. Different but comparable procedures were used for the 3D and Photograph conditions.

In the NASA Johnson Space Center (JSC) Human Factors Engineering Laboratory (HFEL) study, we enhanced the 3D model to include a legend of parts and to show arrow overlays identifying which parts to assemble at each step. Participants rated the visual aids for both HERA C7 and the JSC HFEL lab study using a 5-point Likert scale (1-strongly disagree to 5-strongly agree). Overall, the mean and median ratings were higher for the enhanced 3D model used in the JSC HFEL study than for the original 3D model used in the HERA C7 study, indicating that participants viewed these changes to the 3D model as an improvement.

Procedure Style and Work Autonomy

During HERA Campaign 7, data were collected about the effects of procedure style on crew's sense of autonomy. The VITA study used two styles of procedure in HERA C7. The assembly procedures were highly structured, with sequences of detailed instructions (similar to ISS procedures) accompanied by figures or 3D models for every step. The disassembly procedures were much less structured, with limited details about how to disassemble the rover, similar to those used in HERA C6. Disassembly procedures were performed using both the VITA task assistant on a HoloLens 2 and a web browser on a Tablet. A work autonomy survey [Breaugh, 1985] was administered after every session using both of these procedure types. This survey asks the user to rate the statements about work method autonomy using a 5-point Likert scale (1- strongly disagree to 5-strongly agree).

The effect of Task Type (assembly vs disassembly) and Practice (Sessions 1, 2, 3, and 4) on this work method autonomy rating was analyzed using a 2x4 within-subjects ANOVA. The main effect for Task Type is statistically significant, indicating that the participants' perceived work autonomy was significantly higher for the Disassembly Task than the Assembly Task.

These findings from the work autonomy survey data collected in HERA C7 substantiate the crew observations from HERA C6. Participants found that the less structured disassembly procedures gave them a significantly greater sense of work autonomy than the more structured assembly procedures.

Task Progress:

Participants performing the Disassembly Task have recent device familiarity due to the rover assembly they performed a few days earlier. Research indicates that when users are familiar with either the device or the task using the device, they are less likely to seek out and follow procedural instructions [Eiriksdottir and Catrambone, 2011]. Thus, a Disassembly Task that follows an Assembly Task is an example of a task situation where detailed procedural instruction may not be needed (or heeded) by users.

These less structured procedures are not appropriate for all types of mission work, such as high-risk tasks or tasks requiring skilled performance. Less structured procedures can be effective; however, when the crew member has recent task experience, such as disassembly, not long after assembly seen in HERA C7. Thus, when appropriate to the task, less structured procedures can be used as a mitigation to increase crew's sense of autonomy for long-duration exploration missions.

II. Laboratory Study of VITA Support for Task Transitions

A laboratory study was performed investigating aid from the VITA task assistant for task transitions. The aim of this

study is to determine the best practices to improve crew performance when using the VITA virtual task assistant to aid in task transitions, such as handling the interruption of a manual task by an automated task. In this study, the user is responsible for the manual assembly of the rover gripper. Concurrently, they are responsible for the supervision of an automated task (e.g., the automated startup of a habitat system). Supervisory tasks include monitoring the progress on the startup and intervening in the automated task when automation pauses.

The participant monitors progress on automated tasks using virtual notices displayed in the VITA task assistant. These notices are issued when task milestones are achieved, corresponding to changes in system states or environmental levels.

This study evaluated two conditions: with and without virtual notices from the automated task displayed in the task assistant. The condition With Virtual Notices displayed notices in the HoloLens. The condition Without Virtual Notices did not display notices in the HoloLens or in a Tablet. Instead, the participant monitored automated task progress and the need for intervention by periodically checking the procedure display on the laptop. User intervention with the automation was done for both conditions using PRIDE procedures displayed in a browser on a laptop. Task performance was compared with and without virtual notices in the VITA task assistant. This is a within-subjects study.

The VITA laboratory study had 16 participants. The study was conducted in the Human Factors Engineering Laboratory (HFEL) onsite at JSC using subjects from the Human Test Subject Facility (HTSF). Participants were trained prior to the study session. After training, each participant assembled the gripper, but without responsibility to supervise automation. Then a second gripper assembly was performed, during which the participant was responsible for supervising automation. During the second gripper assembly, participants performed supervisory tasks both with and without virtual notices from the automated task. Sessions were counterbalanced for these notification conditions. Since the annual report was submitted in March 2025, the VITA project has completed the analysis of data from the JSC HFEL lab study.

AR Notices to Supervise Automation

Findings are summarized comparing human performance with AR Notices compared to Tablet Monitoring when supervising automation while also performing a rover assembly.

The JSC HFEL lab study evaluated two conditions for supervision of an automated system concurrent with performing another manual task: AR Notices and Tablet Monitoring. All participants (n = 16) preferred monitoring automation using AR Notices in the HoloLens over watching the automation execute using Tablet Monitoring. Participants reported they felt that monitoring the automation was easier using AR notices (M = 5.94, SD = 0.68, Median = 6) than using Tablet Monitoring (M = 3.94, SD = 1.24, Median = 4). A Wilcoxon signed-rank test shows that AR Notices are significantly easier to use when monitoring automation than Tablet Monitoring.

We expected that the time to respond to an event in the automated system (called response time) would be shorter when using AR Notices than when using Tablet Monitoring. The response times were comparable for both conditions, however. We believe response time for Tablet Monitoring was shorter than expected because users were hyper-vigilant. Additionally, response time for AR Notices was longer than expected because users did not feel a sense of urgency to report immediately. As one participant put it, “I also felt less pressure with the HoloLens. I felt less urgency to log the event, whereas with the scrolling steps, I wanted to log the event right when I saw the step”.

Workload, however, was significantly lower for AR Notices than for Tablet Monitoring. These workload effects are discussed below.

A Wilcoxon signed-rank test was done for median workload of the automation supervision task. Workload for the automation supervision task was significantly less for AR Notices (Median=2) than for Tablet Monitoring.

When supervising the automation using AR notices, participants felt they could focus on the assembly task while responding to notices from the habitat automation in a timely manner. They attributed this to the auditory cue emitted when a new notice was received. Participants heavily relied on the auditory cue to shift their attention to the notice pane when a new notice was received. This reduced the effort needed to monitor the monitor automation. In noisy environments like space habitats, other non-visual modalities may be preferred to cue a new notice.

User-reported reliance on auditory cues to detect incoming notices is consistent with the findings of a pilot study conducted prior to the JSC HFEL Study. In that pilot study, five users monitored for new notices while performing an assembly task. Time to detect the notice was measured for two conditions – with and without an auditory tone when a new notice was received. This was a within-subject study. With no auditory cue, 4 of 5 participants took at least 90 seconds to recognize a new notice. With an auditory cue, all participants recognized a new notice within a few seconds.

When supervising the automation using Tablet Monitoring, most participants felt that the combination of the primary and secondary task was “doable,” but left them with little spare capacity with minor delays. Only one participant directly mentioned that they felt they had enough bandwidth for other tasks if needed.

References 1. Breaugh, J. A. The Measurement of Work Autonomy. Human Relations. Volume 38. Number 6. 1985. Pp. 5551-570. 2. Eiriksdottir, E. and R. Catrambone. Procedural Instructions, Principles, and Examples : How to Structure Instructions for Procedural Tasks to Enhance Performance, Learning, and Transfer. Human Factors. 2011.

Bibliography Type: Description: (Last Updated: 03/25/2025)